Pre-review-book DLBCSM201

Lesson 1 - Introduction to Matrices

1.1 Basic concepts of matrices

Question 1

Level of difficulty: E

Display the following table as a graph.

	North	Middle	South
Product A	120	150	100
Product B	277	259	310

Question 2

Level of difficulty: E

For the following matrix A = $\begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}$, specify the transposed matrix.

Solution question 1

Solution hint: In a matrix, there are no header rows and no margin columns.

Short solution: $\begin{pmatrix} 120 & 150 & 100 \\ 277 & 259 & 310 \end{pmatrix}$

Detailed solution: The numbers are transferred in exactly the same order as in the rows and columns, excluding the header row and margin column: $\begin{pmatrix} 120 & 150 & 100 \\ 277 & 259 & 310 \end{pmatrix}$

Solution question 2

Solution hint: Columns and rows are swapped.

Short solution: $A^T = \begin{pmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{pmatrix}$

Detailed solution: The columns and rows are swapped, $A^T = \begin{pmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{pmatrix}$

Question 3

Level of difficulty: M

Specify one column vector and one row vector each.

Question 4

Level of difficulty: M

Determine the main diagonal for the given matrix. Indicate the type of the matrix. Does the matrix represent a special form? If so, what is its name?

A = $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 8 \\ 0 & 0 & 9 \\ 0 & 0 & 0 \end{pmatrix}$

Solution question 3

Solution hint: Row vectors are to be represented as $\vec{v}^T$.

Short solution: Column vector $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, row vector $\vec{b}^T = (4\ 5\ 6)$

Detailed solution: With the column vector, only one column is represented; with the row vector, only one row is represented. Column vector $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, row vector $\vec{b}^T = (4\ 5\ 6)$.

Solution question 4

Solution hint: Consider the diagonal elements and the values that are above and below the diagonals. The type is mxn.

Short solution: $a_{11} = 1$, $a_{22} = 7$, $a_{33} = 9$, type 4x3, upper triangular matrix

Detailed solution: The diagonal elements are $a_{11} = 1$, $a_{22} = 7$, $a_{33} = 9$. The matrix has four rows and three columns, so the type is 4x3. Since the values are above the diagonals, it is called an upper triangular matrix.

Question 5

Level of difficulty: D

Consider whether the following statements are false:

- If A^T is a diagonal matrix, then A is also a diagonal matrix.
- If A is a zero matrix, then A^T is also a zero matrix.

Solution question 5

Solution hint: Take a close look at the structure of the matrix.

Short solution: Both statements are true.

Detailed solution: Transposing a diagonal matrix preserves the property of a diagonal matrix. If A is a zero matrix, then it contains only zeros. Thus, the transposed matrix is also a zero matrix.

1.2 Addition of matrices

Question 6

Level of difficulty: E

Given the following matrices A and B, calculate A + B.

A = $\begin{pmatrix} -1 & 4 & 6 \\ 2 & -3 & 7 \end{pmatrix}$ B = $\begin{pmatrix} 2 & 3 & 5 \\ 0 & 4 & 7 \end{pmatrix}$

Question 7

Level of difficulty: E

Given the following matrices A and B, is the arithmetic operation A – B possible? If yes, determine the result.

A = $\begin{pmatrix} 4 & 2 & 7 \\ 3 & 1 & 0 \end{pmatrix}$ B = $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$

Solution question 6

Solution hint: Add up the individual elements.

Short solution: C = $\begin{pmatrix} 1 & 7 & 11 \\ 2 & 1 & 14 \end{pmatrix}$

Detailed solution: Add up each of the elements of the matrices, e.g., c_{11} = -1 + 2 = 1.
C = $\begin{pmatrix} 1 & 7 & 11 \\ 2 & 1 & 14 \end{pmatrix}$

Solution question 7

Solution hint: Check the type of the matrices.

Short solution: Arithmetic operation not possible.

Detailed solution: Matrix A is a 2x3 matrix, matrix B is a 2x2 matrix. The matrices must be of the same type in order to add them.

Question 8

Level of difficulty: M

Given the matrices A = $\begin{pmatrix} 1 & 3 & 2 \\ -3 & 4 & 5 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & -1 \\ 0 & 4 \\ 1 & 5 \end{pmatrix}$, calculate A – B^T.

Question 9

Level of difficulty: M

Given the lower triangular matrices below, show that the sum of both is also a lower triangular matrix.

$A = \begin{pmatrix} -1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 4 & 1 \end{pmatrix}$, B = $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

Solution question 8

Solution hint: First determine B^T.

Short solution: A – $B^T = \begin{pmatrix} 1 & 3 & 1 \\ -2 & 0 & 0 \end{pmatrix}$

Detailed solution: B^T = - $\begin{pmatrix} 2 & 0 & 1 \\ -1 & 4 & 5 \end{pmatrix}$. A – B^T = $\begin{pmatrix} 1 & 3 & 2 \\ -3 & 4 & 5 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 1 \\ -1 & 4 & 5 \end{pmatrix}$ = $\begin{pmatrix} -1 & 3 & 1 \\ -2 & 0 & 0 \end{pmatrix}$

Solution question 9

Solution hint: Add both matrices.

Short solution: C = $\begin{pmatrix} 0 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 2 \end{pmatrix}$

Detailed solution: A + B = $\begin{pmatrix} 0 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 2 \end{pmatrix}$. Since only non-zero numbers are below the diagonal, the sum is also a lower triangular matrix.

Question 10

Level of difficulty: D

Given the two diagonal matrices A = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ and B = $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, show that A + B^T also results in a diagonal matrix.

Solution question 10

Solution: First, determine B^T. Then add A and B^T.

Short solution: C = A + B^T = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 9 \end{pmatrix}$

Detailed solution: B^T = $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$. C = A + B^T = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 9 \end{pmatrix}$. The new matrix is also a diagonal matrix.

1.3 Scalar multiplication and product

Question 11

Level of difficulty: E

Given the matrices A = $\begin{pmatrix} 0 & 2 & -1 \\ 1 & 4 & 5 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$, calculate 2 · A + B.

Question 12

Level of difficulty: E

Given the vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$, calculate the scalar product $\vec{a}^{\mathsf{T}} \cdot \vec{b}$.

Solution question 11

Solution hint: First, determine 2 · A.

Short solution: C = $\begin{pmatrix} 2 & 7 & 2 \\ 4 & 9 & 10 \end{pmatrix}$

Detailed solution: 2A + B = 2 · $\begin{pmatrix} 0 & 2 & -1 \\ 1 & 4 & 5 \end{pmatrix}$ + $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$ = $\begin{pmatrix} 0 & 4 & -2 \\ 2 & 8 & 10 \end{pmatrix}$ + $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$ = $\begin{pmatrix} 2 & 7 & 2 \\ 4 & 9 & 10 \end{pmatrix}$

Solution question 12

Solution hint: First, determine $\vec{a}^{\mathsf{T}}$. The result is a scalar.

Short solution: $\vec{a}^{\mathsf{T}} \cdot \vec{b}$ = 23

Detailed solution: $\vec{a}^{\mathsf{T}} \cdot \vec{b}$ = $(1\ 2\ 3) \cdot \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$ = 0 + 8 + 15 = 23

Question 13

Level of difficulty: M

Given the two matrices A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, calculate $2 \cdot A - 3 \cdot B^T$.

Question 14

Level of difficulty: M

Given the vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, and $\vec{c} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, check whether the following arithmetic operations are defined. If so, state the result.

a) $\vec{a} \cdot \vec{b}$ b) $\vec{b} \cdot \vec{c}$ c) $\vec{b}^T \cdot \vec{c}$

Solution question 13

Solution hint: First determine B^T. Then calculate $2 \cdot A$ and $3 \cdot B^T$.

Short solution: $2 \cdot A - 3 \cdot B^T = \begin{pmatrix} -6 & 2 & 4 \\ -7 & -5 & 3 \end{pmatrix}$

Detailed solution: $B^T = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 1 \end{pmatrix}$, $3 \cdot B^T = \begin{pmatrix} 6 & 0 & 0 \\ 3 & 9 & 3 \end{pmatrix}$, $\cdot A = \begin{pmatrix} 0 & 2 & 4 \\ -4 & 4 & 6 \end{pmatrix}$, $2 \cdot A - 3 \cdot B^T = \begin{pmatrix} 0 & 2 & 4 \\ -4 & 4 & 6 \end{pmatrix} - \begin{pmatrix} 6 & 0 & 0 \\ 3 & 9 & 3 \end{pmatrix} = \begin{pmatrix} -6 & 2 & 4 \\ -7 & -5 & 3 \end{pmatrix}$

Solution question 14

Solution hint: Consider the type and definition of the scalar product.

Short answer: a) no b) no c) yes: $\vec{b}^T \cdot \vec{c} = 14$

Detailed solution: a) Both vectors are not of the same type. b) Both vectors are of the same type, but two column vectors cannot be multiplied together. c) Yes: $\vec{b}^T \cdot \vec{c} = (3\ 2) \cdot \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 12 + 2 = 14$

Question 15

Level of difficulty: D

Given the matrix A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$, can the matrix be represented as the sum of a symmetric matrix and a skew-symmetric matrix?

Solution question 15

Solution hint: Consider the type of the matrix.

Short solution: No

Detailed solution: The matrix cannot be represented as the sum of a symmetric and skew-symmetric matrix because it is not square.

Lesson 2 - Inverting Matrices

2.1 Multiplication of matrices

Question 16

Level of difficulty: E

Given the three matrices A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$, B = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, and C = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, check whether the following arithmetic operations are defined. If so, calculate.

a) A · B b) B · C

Question 17

Level of difficulty: E

Given the two matrices A = $\begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$, calculate A · B. Justify why the arithmetic operation is possible.

Solution question 16

Solution hint: Look at the types of matrices.

Short solution: a) possible: C = $\begin{pmatrix} 0 & 5 \\ -4 & 7 \end{pmatrix}$ b) not possible

Detailed solution: A of type 2x3, B of type 3x2, C of type 3x2, number of columns of the first matrix must be equal to the number of rows of the second matrix. Each row is multiplied by each column.

a) possible: C = $\begin{pmatrix} 0 & 5 \\ -4 & 7 \end{pmatrix}$ b) not possible

Solution question 17

Solution hint: Row times column respectively, consider the types of matrices.

Short solution: The number of columns of A matches the number of rows of B:

C = $\begin{pmatrix} 0 & 2 \\ -2 & -2 \end{pmatrix}$

Detailed solution: A has two columns, B has two rows. The number is the same. Multiply the rows by the columns: C = $\begin{pmatrix} 0 & 2 \\ -2 & -2 \end{pmatrix}$

Question 18

Level of difficulty: M

Calculate the dyadic product of the two vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$.

Question 19

Level of difficulty: M

Given two lower triangular matrices A = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, show that C = A · B is also a lower triangular matrix.

Solution question 18

Solution hint: Dyadic product $\vec{a} \cdot \vec{b}^{\mathsf{T}}$, a matrix is formed.

Short solution: $\vec{a} \cdot \vec{b}^{\mathsf{T}} = \begin{pmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 9 & 6 & 3 \end{pmatrix}$

Detailed Solution: $\vec{a} \cdot \vec{b}^{\mathsf{T}} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot (3\ 2\ 1) = \begin{pmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 9 & 6 & 3 \end{pmatrix}$

Solution question 19

Solution hint: Calculate A · B.

Short solution: C = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$

Detailed solution: C = A · B = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$. The resulting matrix is also a lower triangular matrix.

Question 20

Level of difficulty: D

Using the two matrices A = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 0 & 1 \\ 3 & 1 \end{pmatrix}$, show that $B^T \cdot A^T = (A \cdot B)^T$ holds.

Solution question 20

Solution hint: Calculate the left and right sides.

Short solution: $B^T \cdot A^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$, $(A \cdot B)^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$

Detailed solution: $B^T = \begin{pmatrix} 0 & 3 \\ 1 & 1 \end{pmatrix}$, $A^T = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, $B^T \cdot A^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$, $(A \cdot B)^T = \begin{pmatrix} 3 & 3 \\ 9 & 3 \end{pmatrix}^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$,
i.e., $B^T \cdot A^T = (A \cdot B)^T$

2.2 Properties of matrix multiplication

Question 21

Level of difficulty: E

Given two matrices A = $\begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, calculate C = A · B. What conclusion can be drawn from this calculation?

Question 22

Level of difficulty: E

Given two diagonal matrices A = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$, what is C = A · B? What can you conclude from this?

Solution question 21

Solution hint: Which special form does matrix B represent?

Short solution: If a matrix is multiplied by a unit matrix, the matrix does not change. C = A

Detailed solution: C = A · B = $\begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$. C = A · E = A. If a matrix is multiplied by an identity matrix, it remains the same.

Solution question 22

Solution hint: Multiply the two matrices.

Short solution: C = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 12 \end{pmatrix}$. The result is a diagonal matrix.

Detailed solution: C = A · B = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 12 \end{pmatrix}$. The diagonal elements are multiplied. The result is a diagonal matrix.

Question 23

Level of difficulty: M

Does the commutative law apply to matrix multiplication? Explain your statement with the help of an example.

Question 24

Level of difficulty: M

The matrix A = $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 4 & 2 & 1 \end{pmatrix}$ is multiplied by a zero matrix. What type must this zero matrix be? Give an example and the result. What can you conclude from this?

Solution question 23

Solution: The commutative law is A · B = B · A.

Short solution: The commutative law does not apply.

Detailed solution: Choose two matrices A = $\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix}$.

A · B = $\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 10 \\ 5 & 14 \end{pmatrix}$, B · A = $\begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 5 \\ 10 & 4 \end{pmatrix}$, i.e., A · B ≠ B · A.

The commutative law does not apply.

Solution question 24

Solution hint: First, consider what type the zero matrix must be. Then perform the matrix multiplication.

Short solution: The zero matrix must have three rows. The result is a zero matrix that has as many rows as A and as many columns as the zero matrix.

Detailed solution: Since matrix A is of type 3x3, the zero matrix must have three rows; the number of columns does not matter.

Example: 0 = $\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$. A · O = $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 4 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

The result is a zero matrix that has as many rows as A and as many columns as 0.

Question 25

Level of difficulty: D

Two matrices A = $\begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ x & x & 0 \end{pmatrix}$ and B = $\begin{pmatrix} 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{pmatrix}$, each with placeholders *x*, are multiplied.
At which points can C = A · B assume values not equal to zero?

Solution question 25

Solution hint: Perform the matrix multiplication.

Short solution: The following elements of C are not equal to zero: c_{22}, c_{23}, c_{32}, c_{33}

Detailed solution: C = A · B = $\begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ x & x & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & y & y \\ 0 & y & y \end{pmatrix}$. Only the following elements of the matrix C can be non-zero: c_{22}, c_{23}, c_{32}, c_{33}.

2.3 Inverse matrices

Question 26

Level of difficulty: E

Check whether the inverse of the following matrices can be formed.

a) A = $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ b) B = $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

Question 27

Level of difficulty: E

Is the matrix A = $\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ invertible? Justify your answer.

Solution question 26

Solution hint: Consider which conditions must be fulfilled for a matrix to be invertible.

Short solution: a) no b) no

Detailed solution: a) No, because A is a square diagonal matrix. However, not all diagonal values are non-zero. b) No, because only square matrices are invertible.

Solution question 27

Solution hint: Which condition for a 2x2 square matrix must be met for the inverse to exist?

Short solution: det not equal to zero -> invertible

Detailed solution: det = ad – bc = 8 – 3 = 5 $\neq$ 0. Since the determinant is not zero, the inverse exists.

Question 28

Level of difficulty: M

Determine the inverse of the matrix A = $\begin{pmatrix} -2 & 3 \\ 1 & 4 \end{pmatrix}$.

Question 29

Level of difficulty: M

Determine the inverse of the matrix D = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

Solution question 28

Solution hint: Remember the formula that represents how to determine the inverse of a 2x2 square matrix.

Short solution: $A^{-1} = \begin{pmatrix} -\frac{4}{11} & \frac{3}{11} \\ \frac{1}{11} & \frac{2}{11} \end{pmatrix}$

Detailed solution: A = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{-8-3}\begin{pmatrix} 4 & -3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{4}{11} & \frac{3}{11} \\ \frac{1}{11} & \frac{2}{11} \end{pmatrix}$

Solution question 29

Solution hint: Remember the formula that represents how to invert a square diagonal matrix.

Short solution: $D^{-1} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$

Detailed solution: A square matrix is invertible if its diagonal elements are not equal to zero. The inverse is determined by finding the reciprocal of the diagonal elements.

D = $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$, $D^{-1} = \begin{pmatrix} 1/a & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & 1/c \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$

Question 30

Level of difficulty: D

Show by example that if A^{-1} is the inverse of A, then A is the inverse of A^{-1}.

Solution question 30

Solution hint: Choose a simple square matrix and show that $A \cdot A^{-1} = A^{-1} \cdot A$ holds.

Short solution: $A \cdot A^{-1} = A^{-1} \cdot A = E$

Detailed solution: Example: $A = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix}$, $A^{-1} = ½ \begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$,

$A \cdot A^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A^{-1} \cdot A = \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$,

i.e., $A \cdot A^{-1} = A^{-1} \cdot A = E$

Lesson 3 - Linear systems of equations

3.1 Gauss algorithm

Question 31

Level of difficulty: E

Decide whether the following system of equations is linear:

$2x_1 - 3x_2 = 5$

$3(x_1 - 2)^2 + x_2 = x_1$

Question 32

Level of difficulty: E

What are the possible solutions to a system of linear equations?

Solution question 31

Solution hint: Consider the powers of x_1 and x_2.

Short solution: No

Detailed solution: No, because x_1^2 and x_1x_2 are present. For a system of equations to be linear, the variables must be represented only to the first power and not multiplied together.

Solution question 32

Solution hint: There are three possibilities.

Solution: Exactly one solution; infinitely many solutions; no solution

Question 33

Level of difficulty: M

Based on the structure of the augmented matrix $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right)$, what can be said about the solvability of the linear system of equations?

Question 34

Level of difficulty: M

Based on the structure of the following augmented matrix $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 3 \end{array}\right)$, what can be said about the solvability of the linear system of equations?

Solution question 33

Solution hint: Zero lines can be omitted.

Solution: $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & -1 \end{array}\right)$. The third line, i.e., the zero line, can be omitted. After transforming the second line, you get the contradiction 0 = -1. That is, the system of equations is not solvable.

Solution question 34

Solution hint: How many unknowns and how many equations are given?

Short solution: Infinite solutions

Detailed solution: There are three unknowns and only two equations. For the system of equations to be uniquely solvable, there must be as many equations as unknowns.

Question 35

Level of difficulty: D

Give an augmented matrix with four equations and two unknowns such that the associated system of equations is uniquely solvable. Give the solution of the system of equations.

Solution question 35

Solution hint: Two equations should be able to be represented as a combination of the other two.

Short solution: $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \\ 2 & 4 & 6 \end{array}\right)$

Detailed Solution: $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \\ 2 & 4 & 6 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 2 \end{array}\right) \rightarrow$ $x_2 = 2$, $x_1 = -1$; several solutions possible

3.2 Solution examples with the Gauss algorithm

Question 36

Level of difficulty: E

Solve the system of linear equations $\left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right)$

Question 37

Level of difficulty: E

Solve the system of linear equations $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 3 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{array}\right)$

Solution question 36

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 3$, $x_2 = 4$, $x_3 = -5$

Detailed Solution: $\left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -2 & -2 \\ 0 & -1 & -1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -2 & -2 \\ 0 & 0 & 1 & -5 \end{array}\right)$

$\rightarrow x_3 = -5$, $-3x_2 + 10 = -2 \rightarrow x_2 = 4$, $x_1 + 8 - 10 = 1 \rightarrow x_1 = 3$

$\rightarrow$ $x_1 = 3$, $x_2 = 4$, $x_3 = -5$

Solution question 37

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Solution: $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 3 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{array}\right) \rightarrow x_2 = 0$ and $x_2 = -1$. Contradiction! The system of equations is not solvable.

Question 38

Level of difficulty: M

Solve the system of linear equations $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 \end{array}\right)$

Question 39

Level of difficulty: M

Solve the system of linear equations $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 4 & 6 & 1 & 2 \end{array}\right)$

Solution question 38

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 5/7$, $x_2 = -3/7$, $x_3 = 6/7$

Detailed Solution: $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 2 & 1 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 7 & 6 \end{array}\right) \rightarrow x_3 = 6/7$, $x_2 = -3/7$, $x_1 = 5/7$

Solution question 39

Solution hint: Perform the allowed row operations so that an upper triangular matrix is formed.

Short solution: $x_1 = 2$, $x_2 = -1$, $x_3 = 0$

Detailed Solution: $\left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 4 & 6 & 1 & 2 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 0 \end{array}\right) \rightarrow x_3 = 0$, $x_2 = -1$, $x_1 = 2$

Question 40

Level of difficulty: D

Determine the inverse of the matrix A = $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}$.

Solution question 40

Solution hint: Expand the right side with the unit matrix. Reformulate the expanded matrix so that the unit matrix is formed on the left.

Short solution: $A^{-1} = \begin{pmatrix} -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2} \\ +\frac{1}{2} & -\frac{3}{2} & +\frac{1}{2} \\ +\frac{1}{2} & +\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

Detailed solution: $\left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & -1 & -3 & -2 & 0 & 1 \end{array}\right)$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$$

$$\rightarrow A^{-1} = \begin{pmatrix} -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2} \\ +\frac{1}{2} & -\frac{3}{2} & +\frac{1}{2} \\ +\frac{1}{2} & +\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

Lesson 4 - Introduction to Graphs

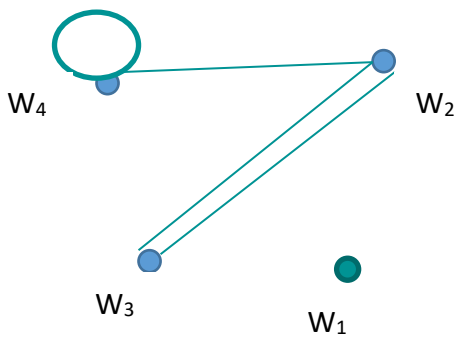
4.1 Undirected graph

Question 41

Level of difficulty: E

Given the graph (right), decide whether the following statements are true or false:

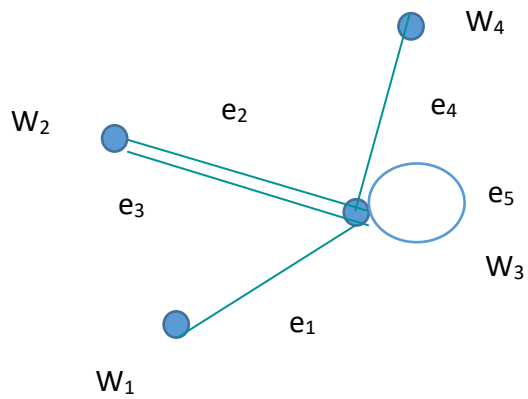
a) The graph is a simple graph.
b) The graph consists of four nodes and four edges.
c) w_3 and w_4 are adjacent.



Question 42

Level of difficulty: E

For the graph (right), give the set of nodes V, the edge set E, and the incidence mapping ϕ.



Solution question 41

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) false, b) true, c) false

Detailed solution: a) False, because the graph has multiple edges and a loop, b) Correct, c) False, because both nodes are not connected by an edge.

Solution question 42

Solution hint: Familiarize yourself with the correct notation.

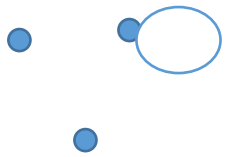
Short solution: $V = \{w_1, w_2, w_3, w_4\}$, $E = \{e_1, e_2, e_3, e_4, e_5\}$, ϕ: $e_1 \to \{w_1, w_3\}$, $e_2 \to \{w_2, w_3\}$, $e_3 \to \{w_2, w_3\}$, $e_4 \to \{w_3, w_4\}$, $e_5 \to \{w_3\}$

Detailed solution: Node set $V = \{w_1, w_2, w_3, w_4\}$, edge set $E = \{e_1, e_2, e_3, e_4, e_5\}$, incidence mapping ϕ: $e_1 \to \{w_1, w_3\}$, $e_2 \to \{w_2, w_3\}$, $e_3 \to \{w_2, w_3\}$, $e_4 \to \{w_3, w_4\}$, $e_5 \to \{w_3\}$

Question 43

Level of difficulty: M

Given the graph (right), decide whether the following statements are true or false.

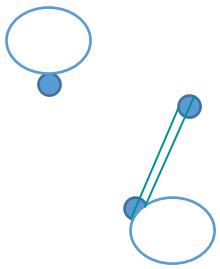


a) The graph is a finite graph.
b) The graph is a simple graph.
c) The graph is a connected graph.

Question 44

Level of difficulty: M

Given the graph (right), decide whether the following statements are true or false.



a) All nodes are adjacent.
b) The graph has a loop.
c) The graph consists of two subgraphs.

Solution question 43

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) true b) true c) false

Detailed solution: a) True, since there are a finite number of nodes and edges, b) False, since it has a loop, c) False, since isolated nodes exist.

Solution question 44

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) false b) true c) true

Detailed solution: a) False, because there are nodes that are not connected by an edge, b) True, because the graph has two loops, c) True, since two graphs are not connected.

Question 45

Level of difficulty: D

Given the graph (right), decide whether the following statements are true or false.

a) The graph is a simple graph.
b) It is not a graph.
c) The graph is contiguous.

Solution question 45

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) true b) false c) true

Detailed solution: a) True, since there are no multiple edges and loops, b) False, since it has one node, c) True, since the graph is not isolated from another graph.

4.2 Further properties of graphs

Question 46

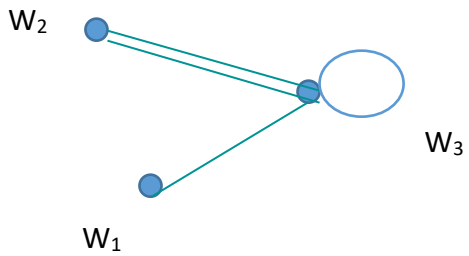
Level of difficulty: E

Draw a graph with four nodes, where one node has degree 1 and three nodes have degree 3.

Question 47

Level of difficulty: E

Determine the degree of each node.

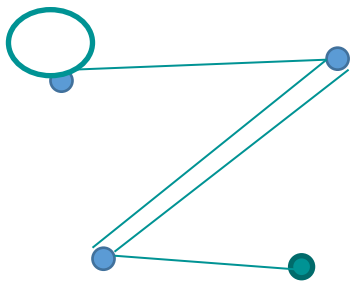


Solution question 46

Solution Note: Use multiple edges and loops if necessary.

Solution: Several solutions are possible

(loop is counted twice).



Solution question 47

Solution hint: The loop is counted twice.

Short solution: Degree of w_1: 1, degree of w_2: 2, degree of w_3: 5

Detailed solution: Degree of w_1: 1, degree of w_2: 2, degree of w_3: 5 (loop is counted twice)

Question 48

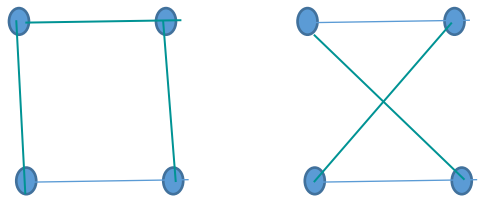
Level of difficulty: M

What do we mean when two graphs are isomorphic? Is it sufficient if they have the same number of nodes and edges?

Question 49

Level of difficulty: M

Are both graphs isomorphic? If yes, give reasons.



Solution question 48

Solution hint: Familiarize yourself with the definition of isomorphism.

Short solution: Two graphs are isomorphic if one graph is formed by redrawing the other graph; no.

Detailed solution: Two graphs are isomorphic if one is formed by redrawing the other. The edges may be bent, stretched, or contracted, but not cut or knotted. For isomorphism, it is not sufficient that two graphs have the same number of nodes, edges, and degrees.

Solution question 49

Solution hint: Familiarize yourself with the definition of isomorphism.

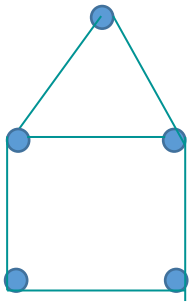
Short solution: Yes; label the edges and nodes

Detailed solution: The graphs are isomorphic. One possibility is labeling the nodes and edges so that isomorphism can be seen.

Question 50

Level of difficulty: D

Verify Euler's formula for the following graph.



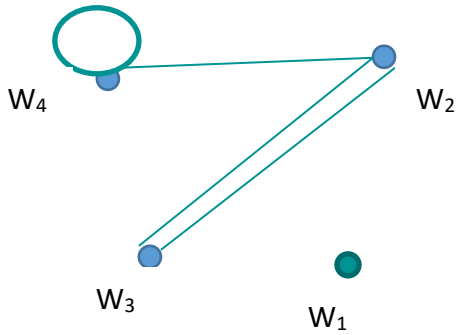
Answer question 50

Solution hint: The area outside the graph is counted.

Short solution: v – e + f = 2; 5 – 6 + 3 = 2

Detailed solution: Number of nodes v – number of edges e + number of faces f = 5 – 6 + 3 = 2. The result corresponds to Euler's formula.

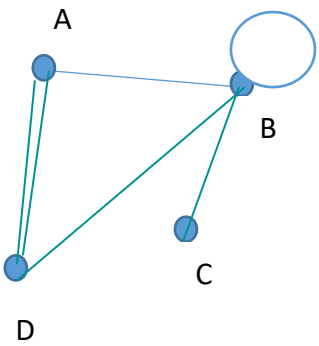
4.3 Adjacency matrix



Question 51

Level of difficulty: E

Given the following graph, state the adjacency matrix.



Question 52

Level of difficulty: E

Given the following graph, state the adjacency matrix.

Solution question 51

Solution hint: Loops are counted twice.

Solution: $A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix}$

Solution question 52

Solution hint: Loops are counted twice.

Solution: $A = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix}$

Question 53

Level of difficulty: M

A graph is given by the following adjacency matrix. Make a drawing.

$$A = \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Question 54

Level of difficulty: M

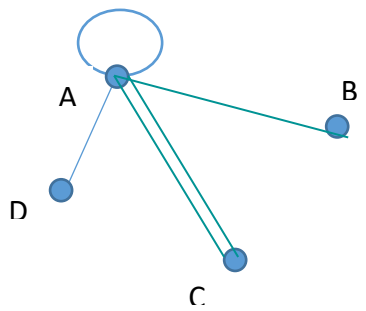
A graph is given by the following adjacency matrix. Make a drawing.

$$A = \begin{pmatrix} 2 & 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution question 53

Solution hint: First draw the nodes A, B, C and D.

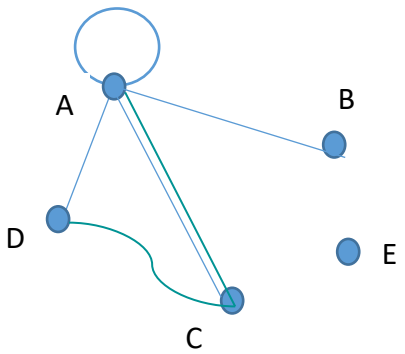
Solution: See graph



Solution question 54

Solution hint: First draw the nodes A, B, C, D and E.

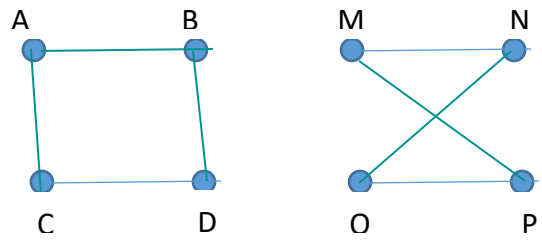
Solution: See graph



Question 55

Level of difficulty: D

Check against the adjacency matrix, whether the graphs are isomorphic to each other.



Solution question 55

Solution hint: To do this, give a bijective mapping of the set of nodes of G to the set of nodes of H and check that the two adjacency matrices are equal.

Solution: The two graphs are isomorphic. A -> M, B -> N, C -> P, D -> O

$A_{left} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$ $A_{right} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$

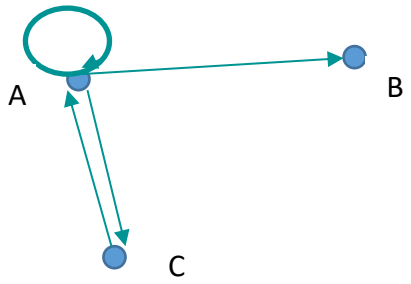
Lesson 5 - Shortest Path Problem

5.1 Directed graph or digraph

Question 56

Level of difficulty: E

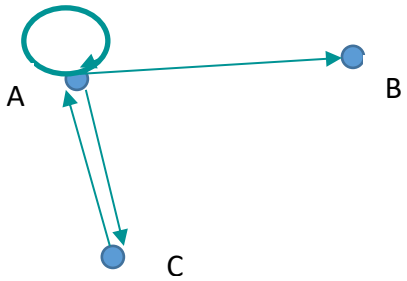
Given the digraph, enter the input and output degrees for all nodes.



Question 57

Level of difficulty: E

Given the digraph, draw its shadow.



Solution question 56

Solution hint: Look up how many edges (arrows) lead in and out of the respective nodes.

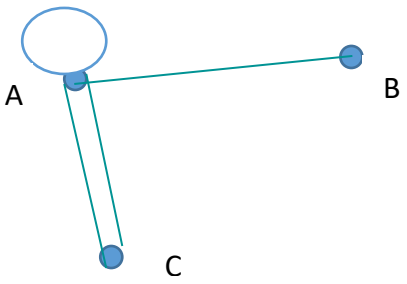
Solution:

	Input degree	Output degree
A	2	3
B	1	0
C	1	1

Solution question 57

Solution hint: The shadow of a digraph is an undirected graph.

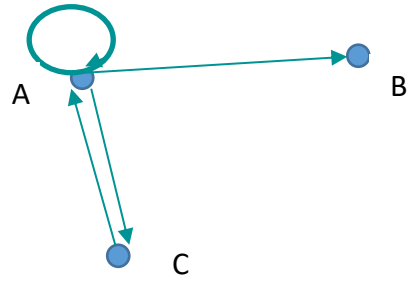
Solution:



Question 58

Level of difficulty: M

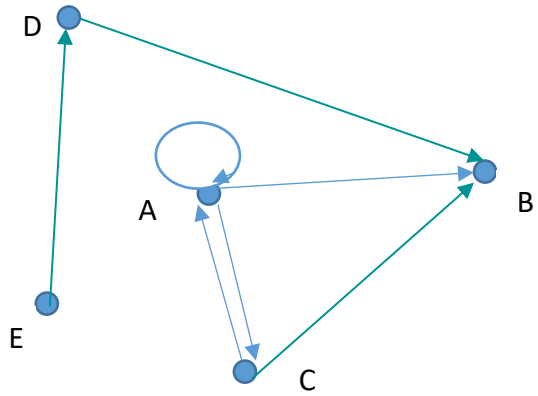
Specify the adjacency matrix of the given digraph.



Question 59

Level of difficulty: M

Specify the adjacency matrix of the given digraph.



Solution question 58

Solution hint: The adjacency matrix specifies how many directed edges from one node enter each of the other nodes.

Solution: A = $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

Solution question 59

Solution hint: The adjacency matrix specifies how many directed edges from one node enter each of the other nodes.

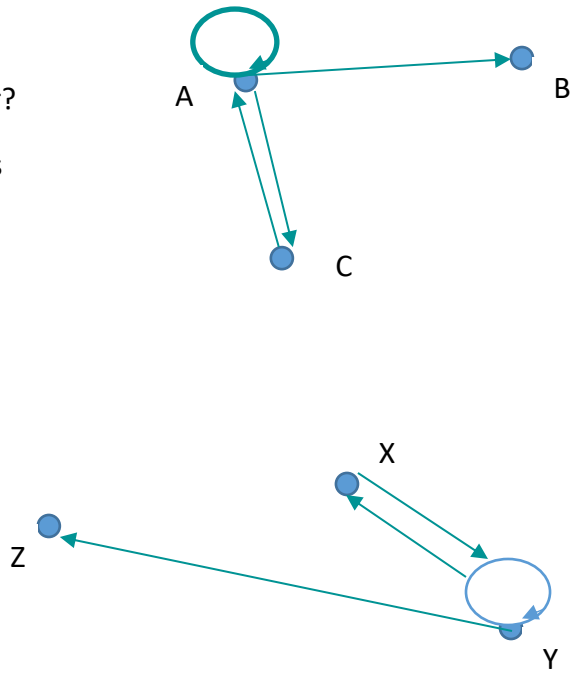
Solution: A = $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

Question 60

Level of difficulty: D

Are the two graphs given isomorphic to each other?

If yes, state the bijective mapping that both graphs transfer into each other.



Solution question 60

Solution hint: Give a bijective mapping of the set of nodes of the first graph to the set of nodes of the second graph.

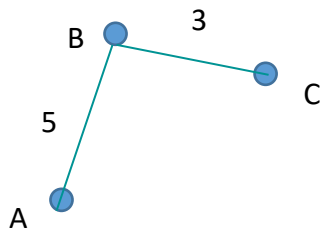
Solution: The two graphs are isomorphic. A -> Y, B -> Z, C -> X

5.2 Weighted graph

Question 61

Level of difficulty: E

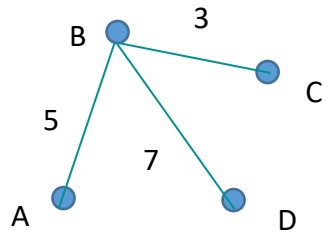
For the following graph, give the weight matrix C.



Question 62

Level of difficulty: E

For the following graph, give the weight matrix C.



Solution question 61

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & 5 & \infty \\ 5 & 0 & 3 \\ \infty & 3 & 0 \end{pmatrix}$

Solution question 62

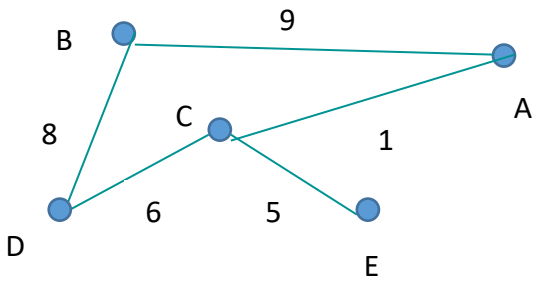
Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & 5 & \infty & \infty \\ 5 & 0 & 3 & 7 \\ \infty & 3 & 0 & \infty \\ \infty & 7 & \infty & 0 \end{pmatrix}$

Question 63

Level of difficulty: M

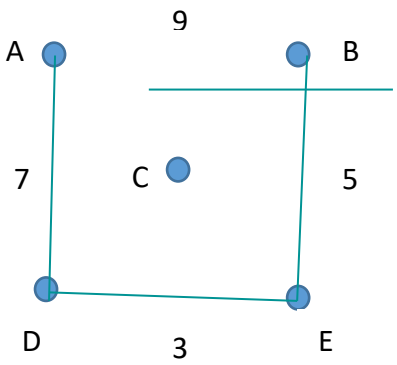
For the following graph, give the weight matrix C.



Question 64

Level of difficulty: M

For the following graph, give the weight matrix C.



Solution question 63

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: $C = \begin{pmatrix} 0 & 9 & 1 & \infty & \infty \\ 9 & 0 & \infty & 8 & \infty \\ 1 & \infty & 0 & 6 & 5 \\ \infty & 8 & 6 & 0 & \infty \\ \infty & \infty & 5 & \infty & 0 \end{pmatrix}$

Solution question 64

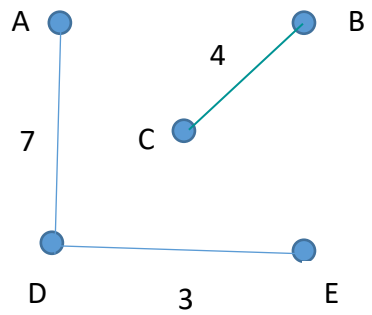
Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: $C = \begin{pmatrix} 0 & 9 & \infty & 7 & \infty \\ 9 & 0 & \infty & \infty & 5 \\ \infty & \infty & 0 & \infty & \infty \\ 7 & \infty & \infty & 0 & 3 \\ \infty & 5 & \infty & 3 & 0 \end{pmatrix}$

Question 65

Level of difficulty: S

For the following graph, give the weight matrix C.



Solution question 65

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & \infty & \infty & 7 & \infty \\ \infty & 0 & 4 & \infty & \infty \\ \infty & 4 & 0 & \infty & \infty \\ 7 & \infty & \infty & 0 & 3 \\ \infty & \infty & \infty & 3 & 0 \end{pmatrix}$

5.3 Dijkstra's algorithm

Question 66

Level of difficulty: E

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 4 & 5 \\ \infty & 0 & 2 \\ 3 & 1 & 0 \end{pmatrix}$, where the edges are sorted alphabetically, determine the shortest paths from the starting node A to all other nodes.

Question 67

Level of difficulty: E

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 6 \\ 1 & 0 & \infty \\ \infty & 3 & 0 \end{pmatrix}$, where the edges are sorted alphabetically, determine the shortest paths from the starting node A to all other nodes.

Solution question 66

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	4
C	A -> C	5

Node	1	2
A		
B	**4** A	
C	5 A	**5** A
M	A, B	A, B, C
P	C	

Solution question 67

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	5
C	A -> C	6

Node	1	2
A		
B	**5** A	
C	6 A	**6** A
M	A, B	A, B, C
P	C	

Question 68

Level of difficulty: M

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 7 & 8 \\ 2 & 0 & 1 & 4 \\ 3 & \infty & 0 & 1 \\ 4 & 3 & \infty & 0 \end{pmatrix}$, where the edges are sorted alphabetically: A, B, C, D. Determine the shortest paths from the starting node A to all other nodes.

Question 69

Level of difficulty: M

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 3 & \infty \\ \infty & 0 & 2 & \infty \\ \infty & 1 & 0 & 7 \\ 4 & \infty & \infty & 0 \end{pmatrix}$, where the edges are sorted alphabetically: A, B, C, D. Determine the shortest paths from the starting node A to all other nodes.

Solution question 68

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	5
C	A -> B -> C	6
D	A -> B -> C -> D	7

Node	1	2	3
A			
B	**5** A		
C	7 A	**6** B	
D	8 A	8 A	**7** C
M	A, B	A, B, C	A, B, C, D
P	C, D	D	

Solution question 69

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> C -> B	4
C	A -> C	3
D	A -> C -> D	10

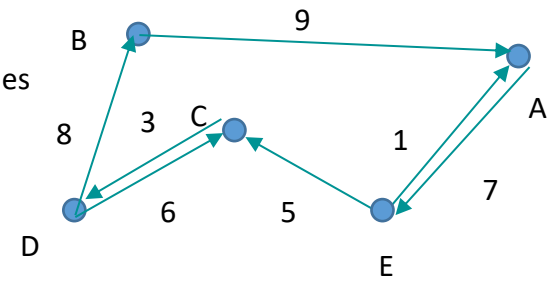
Node	1	2	3
A			
B	5 A	**4** C	
C	**3** A		
D	∞	10 C	**10** C
M	A, C	A, C, B	A, C, B, C
P	B	D	

Question 70

Level of difficulty: D

Given the following weighted digraph, arrange the nodes in alphabetical order.

Using the Dijkstra's algorithm determine the shortest paths from node A to all other nodes.



Solution question 70

Solution hint: Set up the weight matrix C and use Dijkstra's algorithm.

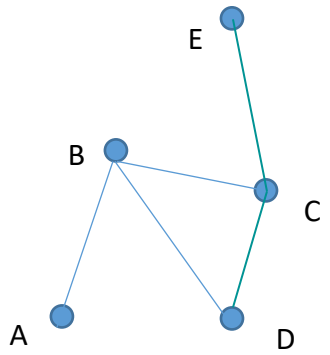
Solution: $C = \begin{pmatrix} 0 & \infty & \infty & \infty & 7 \\ 9 & 0 & \infty & \infty & \infty \\ \infty & \infty & 0 & 3 & \infty \\ \infty & 8 & 6 & 0 & \infty \\ 1 & \infty & 5 & \infty & 0 \end{pmatrix}$

Node	Shortest path	Shortest distance
B	A -> E -> C -> D -> B	23
C	A -> E -> C	12
D	A -> E -> C -> D	15
E	A -> E	7

	1	2	3	4
A				
B	∞			**23** D
C	∞	**12** E		
D	∞		**15** C	
E	**7** A			
M	A, E	A, E, C	A, E, C, D	A, E, C, D, B
P				

Lesson 6 - The Königsberg Bridge Problem

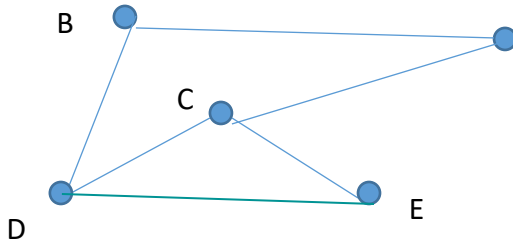
6.1 Walk



Question 71

Level of difficulty: E

Decide whether it is an open or closed walk and state the length in each case: a) ABCE b) BDCB c) BDCE



Question 72

Level of difficulty: E

Decide whether it is an open or closed walk and specify the length in each case:
a) DBAC b) CEDCABDC c) DBACD

Solution question 71

Solution hint: Count the number of edges and check if the walk returns to the initial node.

Solution: a) open walk of length 3, b) closed walk of length 3, c) open walk of length 3

Solution question 72

Solution hint: Count the number of edges and check if the path returns to the initial node.

Solution: a) open walk of length 3, b) closed walk of length 7, c) closed walk of length 4

Question 73

Level of difficulty: M

The following walk belongs to a graph: DCAD closed walk of length 3, ABDCA closed walk of length 4, CADB open walk of length 3. Draw a possible graph for this purpose.

Question 74

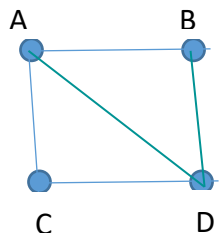
Level of difficulty: M

The following walk belong to a graph: BACD open walk of length 3, ACDA closed walk of length 3, CADCAB open walk of length 5. Draw a possible graph for this purpose.

Solution question 73

Solution hint: Draw the nodes first.

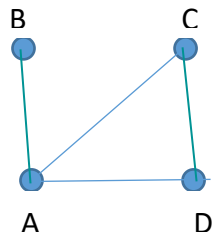
Solution: Several solutions are possible.



Solution question 74

Solution hint: Draw the nodes first.

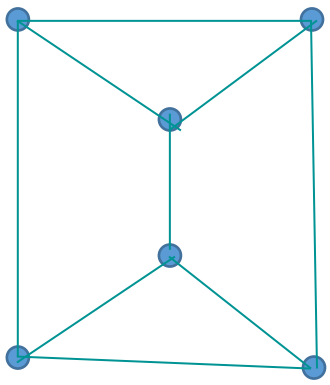
Solution: Several solutions are possible.



Question 75

Level of difficulty: D

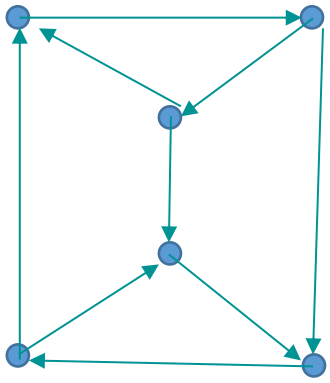
Show that the following graph leads to a strongly connected digraph by giving all edges suitable directions.



Solution question 75

Solution hint: Look up what is meant by a strongly connected digraph.

Solution:

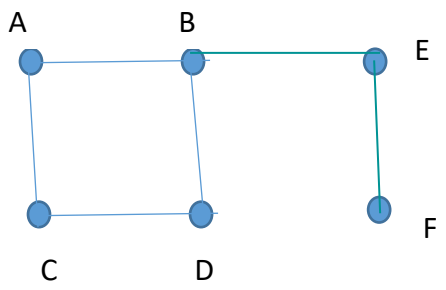


6.2 Eulerian graph

Question 76

Level of difficulty: E

Add an edge to the following graph so that it becomes Eulerian.



Question 77

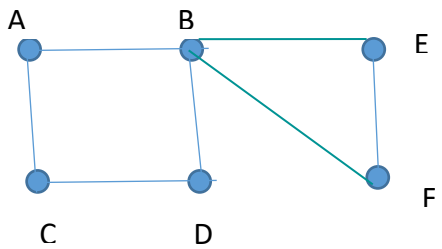
Level of difficulty: E

Is the Nicholas house an Eulerian graph? Justify your answer.

Solution question 76

Solution hint: If the two nodes with odd degree are connected by an additional edge, it is possible to return to the starting point and to draw one or more Eulerian cycles.

Solution: So that all node degrees are straight, an additional edge must be inserted between nodes B and F.



Solution question 77

Solution hint: All node degrees must be even.

Short solution: No

Detailed solution: No, because two nodes have an odd degree.

Question 78

Level of difficulty: M

Is the wheel graph W_5 an Eulerian path, an Eulerian cycle, or neither?

Question 79

Level of difficulty: M

Is the complete graph K_5 an Eulerian cycle? Justify your answer.

Solution question 78

Solution hint: Draw W_5. Which conditions must be met for an Eulerian path or an Eulerian cycle to exist?

Short solution: No

Detailed solution: The wheel graph W_5 is neither an Eulerian path nor an Eulerian cycle. It has four nodes with odd degree 3. Prerequisite for an Eulerian path are max. two nodes with odd degree. The prerequisite for an Eulerian cycle is only nodes with even degree.

Solution question 79

Solution hint: Draw the complete graph. Which conditions must be met for an Eulerian cycle to exist?

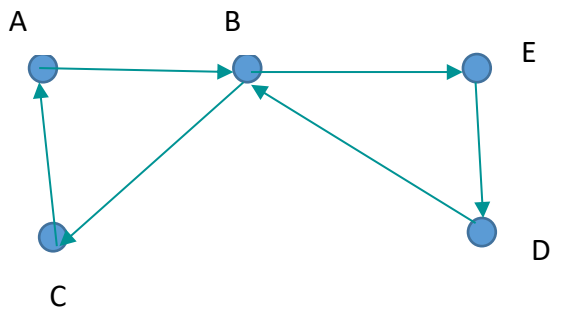
Short solution: Yes

Detailed solution: Yes, since it has only nodes with even node degree. All complete graphs with n odd are Eulerian.

Question 80

Level of difficulty: D

Is the following digraph Eulerian? Give reasons for your answer.



Solution question 80

Solution hint: A digraph is Eulerian if and only if it is strongly connected and for each node it holds that the input degree is equal to the output degree.

Short solution: Yes

Detailed solution: Yes, the digraph is strongly connected and for each node the input and output degrees match.

6.3 Hierholzer's algorithm

Question 81

Level of difficulty: E

What is Hierholzer's algorithm used for? Explain the procedure of this algorithm.

Question 82

Level of difficulty: E

What else is Hierholzer's algorithm called? Why is it called that?

Solution question 81

Solution hint: Eulerian cycle

Solution: Hierholzer's algorithm is used to determine Eulerian cycles in an undirected graph. Prerequisites: Connected graph that has only nodes with even degree.

1. Choose a node and construct from it a subcycle K that does not pass through any edge twice.
2. If K is an Eulerian cycle, then break off. Otherwise, continue to step 3.
3. Neglect all edges of K.
4. At the first vertex of K, let a second subcycle K_2 arise which contains no edge of K and no edge of the original graph twice.
5. Insert the second circle K_2 into K.
6. Continue with step 2.

Solution question 82

Solution hint: Familiarize yourself with the algorithm's procedure.

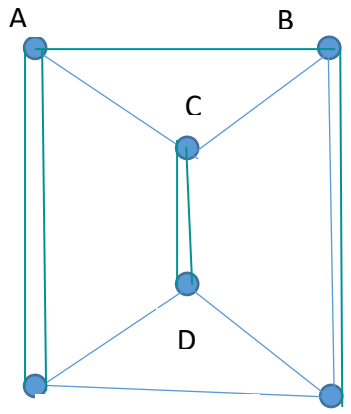
Short solution: Onion skin algorithm

Detailed solution: Onion skin algorithm. The constructed closed walk resembles nested onion skins.

Question 83

Level of difficulty: M

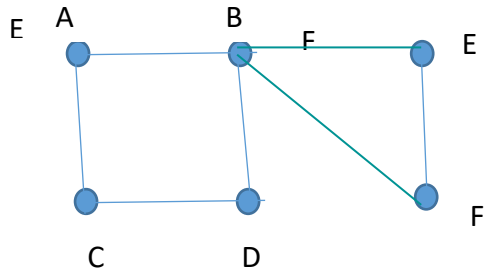
First check whether the given undirected graph is Eulerian. If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Question 84

Level of difficulty: M

First check whether the given graph Eulerian.
If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Solution question 83

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

Short solution: Yes, AEDFBACBFEA

Detailed solution: The given graph is Eulerian since it contains only nodes with even node degree. Hierholzer's algorithm yields an Eulerian cycle AEDFBACBFEA. Multiple solutions are possible.

Solution question 84

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

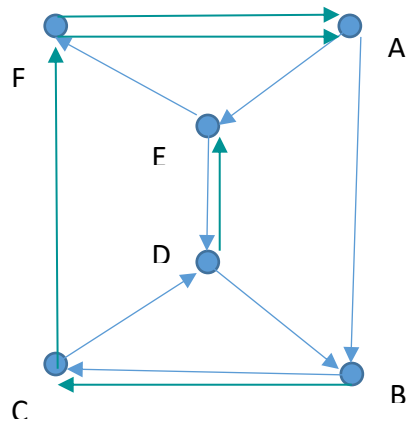
Short solution: Yes, ABEFBDCA

Detailed solution: The given graph is Eulerian since it contains only nodes with even node degree. Hierholzer's algorithm yields an Eulerian cycle ABEFBDCA. Several solutions are possible.

Question 85

Level of difficulty: D

First check whether the given directed graph is Eulerian. If so, then run Hierholzer's algorithm and determine an Eulerian cycle.



Solution question 85

Solution hint: Which conditions must be given for an Eulerian cycle? Look at the procedure for Hierholzer's algorithm.

Short solution: Yes, ABCDBCFAEDEFA

Detailed solution: The given graph is Eulerian since it is strongly connected and at each node the input degree coincides with the output degree. Hierholzer's algorithm yields an Eulerian cycle ABCDBCFAEDEFA.

6.4 The Postman Problem

Question 86

Level of difficulty: E

Explain the postman problem. Give two concrete examples from practice.

Question 87

Level of difficulty: E

Explain in simplified terms the approach to the postman problem.

Solution question 86

Solution hint: Put yourself in the shoes of a postman.

Short solution: Walk all streets with minimal effort and return to starting point; examples include delivery services and street cleaning.

Detailed solution: A postman has to divide accurately his working time. His delivery area comprises several streets through which he must pass at least once. Detours or return routes, if they cannot be avoided, should be as short as possible. The streets are considered edges and the intersections and junctions are considered nodes. The graph is weighted, since the length of the streets plays a role. We are looking for a closed set of edges that contains all streets of the delivery area and has a minimum total weight. Similar problems are generally found in collection and delivery services, such as garbage collection, street cleaning, or factory traffic, but also in city tours, etc.

Solution question 87

Solution hint: Find the shortest walk between every two nodes with odd degree.

Solution: The shortest walk between every two nodes with odd degree, i.e., every pair, is to be found by applying Dijkstra's algorithm. The edges of this walk are to be doubled. The newly created graph is Eulerian, so that an Eulerian cycle is to be found in it using Hierholzer's algorithm.

Question 88

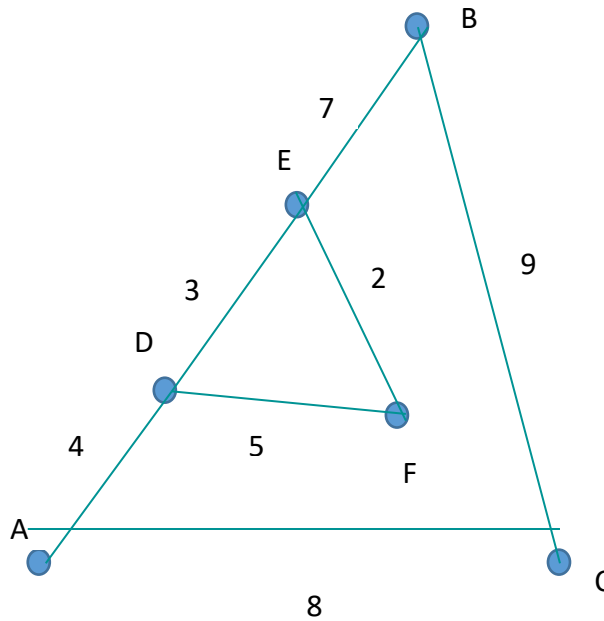
Level of difficulty: M

Explain the solution of the postman problem using graph theory.

Question 89

Level of difficulty: M

For the following graph, give the optimal solution to the postman problem.



Solution question 88

Solution hint: The shortest route corresponds to an Eulerian cycle.

Short solution: An Eulerian cycle is optimal. If the graph is not Eulerian, it must be made Eulerian by adding edges.

Detailed solution: In Eulerian graphs, the shortest route corresponds to an Eulerian cycle. By definition, an Eulerian cycle contains each edge exactly once. However, the problem requires that each edge be contained at least once. Therefore, any solution that contains each edge exactly once is optimal in every case. If the graph is not Eulerian, the problem is solved by making the graph Eulerian by adding optimal edges. For the postman, duplicated edges mean that these roads must be walked twice. If the graph is Eulerian, we proceed as above and we look for an Eulerian cycle as the optimal solution. For a connected graph that is not Eulerian, there are an even number of nodes that have odd degree. If each two nodes of odd degree are connected by an additional shortest walk doubling all edges in this walk, the formerly odd node degrees become even. Nodes with already even node degree within the walk remain even by the doubling. The shortest walk between two nodes of odd degree can be determined using Dijkstra's algorithm.

Solution question 89

Solution hint: Double the edges with odd node degree.

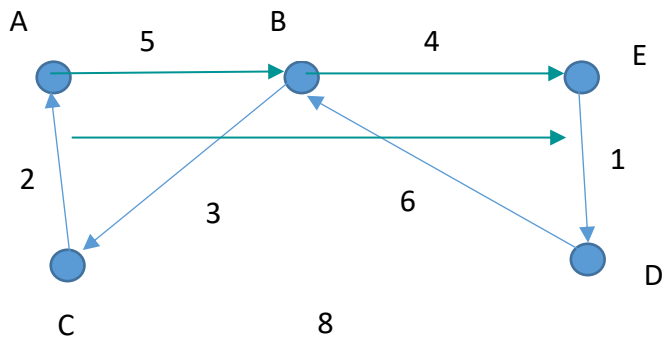
Short solution: Double the edge between D and E; Eulerian cycle: ADEFDEBCA

Detailed solution: D and E are nodes with odd degree. The shortest edge from D to E is the direct one. Thus, this edge is to be doubled. Hierholzer's algorithm yields the following Eulerian cycle: ADEFDEBCA. This is a minimal postman path. Note: The Eulerian cycle need not be unique.

Question 90

Level of difficulty: D

For the following digraph, enter a minimum cost solution for the postman problem.



Solution question 90

Solution hint: Input and output degree must match.

Short solution: Additional arrows are (D, B) and (B, C); Eulerian digraph ABCDBEDBCA.

Detailed solution: At nodes C and D, the input and output degrees do not match. At C another arrow should arrive, at D another arrow should leave. So the shortest directed walk from D to C is searched. This leads over B with length 9. So the additional arrows are (D, B) and (B, C). An Eulerian cycle in the modified Eulerian digraph is ABCDBEDBCA.

Lesson 7 - A City Tour Where Each City Is Visited Exactly Once

7.1 Special graphs

Question 91

Level of difficulty: E

What is meant by a star graph? How many nodes and edges does the star graph S_n have? Draw the star graph S_4.

Question 92

Level of difficulty: E

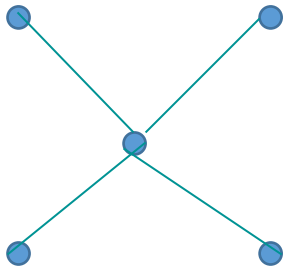
What is meant by cycle graphs? What properties do they have?

Solution question 91

Solution hint: Familiarize yourself with the theory of star graphs.

Short solution: Central node connected to all other nodes; n edges and n+1 nodes

Detailed solution: A star graph S_n has a central node, which is connected to all other nodes by an edge. The other nodes have no neighbors other than the central node.
A star graph S_n has n edges and therefore n+1 nodes.



Solution question 92

Solution hint: Familiarize yourself with the theory of cycle graphs.

Short solution: Same number of nodes as edges, regular of degree 2, planar and Eulerian

Detailed solution: A cycle graph C_n with n (in general, $n \geq 3$ is assumed) vertices has as many vertices as edges, all vertices being connected in a (geometric) cycle. All cycle graphs are regular of degree 2, planar and Eulerian.

Question 93

Level of difficulty: M

What is meant by a complete graph? Are all complete graphs Eulerian? How many edges does a complete graph K_n have?

Question 94

Level of difficulty: M

What is meant by a bipartite graph? What properties must it have? Give an example.

Solution question 93

Solution hint: Familiarize yourself with the theory of complete graphs.

Short solution: From each node, an edge leads to every other node. Not all complete graphs are Eulerian. Number of edges: n * (n–1)/2.

Detailed solution: An undirected simple graph with n vertices in which there exists an edge from every vertex to every other vertex is called a complete graph (or complete polygon) K_n. All complete graphs K_n with n odd are Eulerian. The number of all edges of a complete graph is the triangular number n * (n–1)/2.

Solution question 94

Solution Note: Familiarize yourself with the theory of bipartite graphs.

Short solution: The set of its nodes can be split into two disjoint subsets. Prerequisite: At least two nodes and no loop. Example: Cycle graph with even number of nodes

Detailed solution: An undirected graph is bipartite if the set of its nodes can be split into two disjoint non-empty subsets M and N, such that there are only edges between nodes from M and nodes from N, but no edges between nodes within the set M or the set N. A graph whose edge set is the union of two non-empty disjoint sets M and N, such that each edge has one end in M and one end in N, is a bipartite graph. Because of this definition, it is necessary for a bipartite graph to have at least two nodes and no loops (but it can have multiple edges). In other words, a graph is bipartite if its nodes can be colored black and white such that each edge connects a black and a white node (colorable with two colors). The two subsets M and N are also called bipartite. All cycle graphs C_n with even number of nodes are bipartite.

Question 95

Level of difficulty: D

Explain Kuratowski's theorem. What becomes evident on the basis of this theorem?

Solution question 95

Solution hint: Familiarize yourself with the theory. What does planar mean?

Short solution: A graph G is planar if and only if G does not contain a subgraph G' such that K_5 or $K_{3,3}$ is a minor of G'. Due to Kuratowski's theorem, it is obvious that K_5 and $K_{3,3}$ are the smallest non-planar graphs.

Detailed solution: A graph G_1 is called minor of graph G_2 if G_1 is isomorphic to a subgraph of G_2 formed by node merging. Node merging means that two adjacent nodes are merged into one node by removing one edge connecting these two nodes, while keeping all remaining edges. Using Minor's definition, Kuratowski's theorem can be formulated as follows: A graph G is planar if and only if G does not contain a subgraph G' such that K_5 or $K_{3,3}$ is a minor of G'. Given Kuratowski's theorem, it is evident that K_5 and $K_{,3}$ are the smallest non-planar graphs.

7.2 Hamiltonian graph

Question 96

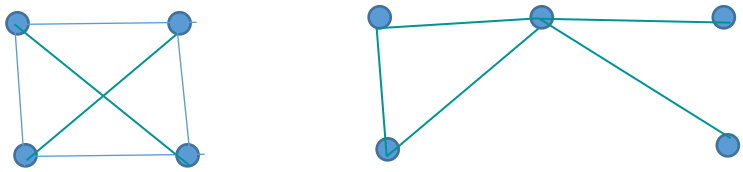
Level of difficulty: E

What is meant by a Hamiltonian path and a Hamiltonian cycle?

Question 97

Level of difficulty: E

Do the following graphs possess a Hamiltonian path and a Hamiltonian cycle?



Solution question 96

Solution hint: Familiarize yourself with the theory of Hamiltonian graphs.

Short solution: The path contains each node only once. Path: not closed. Cycle/circuit: closed

Detailed solution: If an edge path contains each node of a graph exactly once, but is not necessarily closed, then it is a Hamiltonian path. For a directed graph, it is also called a directed Hamiltonian path. A graph that contains a Hamiltonian path is called a semi-Hamiltonian graph. In particular, a Hamiltonian cycle is a closed Hamiltonian path.

Solution question 97

Solution hint: The path contains each node only once.

Short solution: Graph on the left: Hamiltonian path and Hamiltonian cycle; Graph right: no

Detailed solution: Graph on the left: Hamiltonian path and Hamiltonian cycle. There exists a closed path that contains each node only once. Graph on the right: There exists neither an open nor a closed path, which contains each node only once.

Question 98

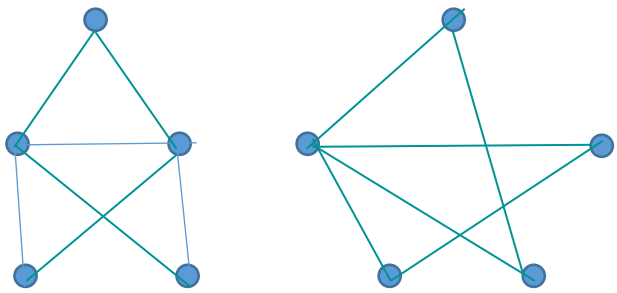
Level of difficulty: M

Are path graphs, cycle graphs, star graphs, and wheel graphs Eulerian and/or Hamiltonian?

Question 99

Level of difficulty: M

Do the following graphs possess a Hamiltonian path and a Hamiltonian cycle?



Solution question 98

Solution hint: What does Eulerian and Hamiltonian mean?

Solution: Path graphs and star graphs are not Eulerian and not Hamiltonian; cycle graphs are Eulerian and Hamiltonian; wheel graphs ($n \geq 4$) are not Eulerian but Hamiltonian.

Solution question 99

Solution hint: The path contains each node only once.

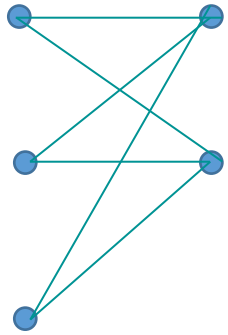
Short solution: Both graphs have a Hamiltonian path.

Detailed solution: In both graphs, there exists an open path that contains each node only once, i.e., an Eulerian path. However, the path cannot be closed.

Question 100

Level of difficulty: D

Show that the graph is not Hamiltonian. Use the result in reference to bipartite graphs.



Solution question 100

Solution hint: If a graph is Hamiltonian with more than two nodes and bipartite, then the two sets of nodes that form the bipartite partitioning are equipotent.

Short solution: Disjoint subsets are not equally powerful.

Detailed solution: The graph is bipartite. Since the disjoint subsets contain three and two nodes and are therefore not equal, the graph cannot be Hamiltonian.

7.3 The Ore and Dirac conditions

Question 101

Level of difficulty: E

Is a graph non-Hamiltonian if it does not satisfy the Ore condition? Give an example.

Question 102

Level of difficulty: E

If a graph satisfies the Dirac condition, does it also satisfy the Ore condition? If a graph satisfies the Ore condition, does it also satisfy the Dirac condition?

Solution question 101

Solution hint: Familiarize yourself with the Ore condition.

Short solution: It does not mean that the graph is not Hamiltonian. Example: C_5

Detailed solution: If a simple graph does not satisfy the Ore condition, this does not mean that it is not a Hamiltonian graph. Thus, C_5 is a simple Hamiltonian graph, but for every two nonadjoint vertices u and v, it holds that $\delta(u) + \delta(v) = 2 + 2 \ngeq n = 5$. This means that the cycle graph C_5 does not satisfy the Ore condition.

Solution question 102

Solution hint: Familiarize yourself with the Ore condition and Dirac condition.

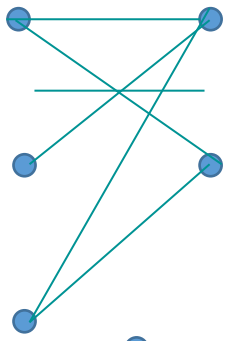
Short solution: Yes; no

Detailed solution: If a graph satisfies the Dirac condition, then it also satisfies the Ore condition. However, if a graph satisfies the Ore condition, then it need not also satisfy the Dirac condition.

Question 103

Level of difficulty: M

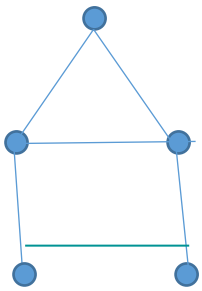
Apply the Ore and Dirac condition to the following graphs. What can you conclude?



Question 104

Level of difficulty: M

Apply the Ore and Dirac condition to the following graphs. What can you conclude?



Solution question 103

Solution hint: Familiarize yourself with the two conditions.

Short solution: Both equations were not fulfilled. Thus, no decision can be made.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 3 = 6$, respectively $\delta(u) + \delta(v) = 2 + 2 = 4 \ngeq 5$. In other words, the Ore condition does not hold for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 2 \ngeq 5/2$. In other words, the Dirac condition is not satisfied. It follows that the Ore and Dirac conditions cannot be used to decide whether the graph is Hamiltonian.

Solution question 104

Solution hint: Familiarize yourself with the two conditions.

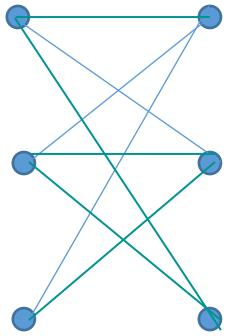
Short solution: Both equations were not fulfilled. Thus, no decision can be made.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 2 = 5$ and $\delta(u) + \delta(v) = 2 + 2 = 4 \ngeq 5$. In other words, the Ore condition does not hold for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 2 \ngeq 5/2$. In other words, the Dirac condition is not satisfied. It follows that the Ore and Dirac conditions cannot be used to decide whether the graph is Hamiltonian. Without proof, let us mention that the graph is Hamiltonian.

Question 105

Level of difficulty: D

Apply the Ore and Dirac condition to the following graph. What can you conclude from this?



Solution question 105

Solution hint: Familiarize yourself with the conditions.

Short solution: Both conditions are satisfied. The graph is Hamiltonian.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 3 = 6 \geq 6$. In other words, the Ore condition is satisfied for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 3 \geq 6/2$. In other words, the Dirac condition is satisfied. It follows that the graph is Hamiltonian.

7.4 Problem of the traveling salesman

Question 106

Level of difficulty: E

What is understood by the traveling salesman problem? Where does it find application in practice? Give an example.

Question 107

Level of difficulty: E

What is the problem with the approach to the problem of the traveling salesman?

Solution question 106

Solution hint: Hamilton cycle problem for weighted graphs

Short solution: Round trip through all nodes minimizing the sum of the weights. Example: electronic circuit boards.

Detailed solution: Closely related to the Hamilton cycle problem is the traveling salesman problem (or TSP). It differs from the Hamilton circle problem only in that the underlying graphs are weighted. The task is to find a round trip through all nodes that minimizes the sum of the distances traveled. So the task is to find a Hamiltonian cycle in a weighted graph such that the sum of the weights is minimal. Example: Electronic circuit board manufacturing.

Solution question 107

Solution hint: Heuristics

Short solution: There is no exact solution.

Detailed solution: There is no exact algorithm that can solve the problem in acceptable time. Of course, it is possible to determine all Hamiltonian cycles, compare the sum of the weights of each, and choose the cycle with the smallest weight. However, as described earlier, a simple graph with n nodes can have up to $(n-1)!/2$ Hamiltonian cycles. This is a lot of possibilities to compare.

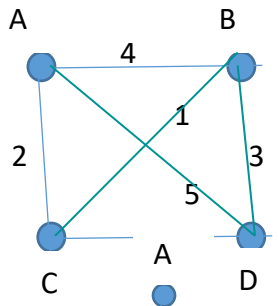
Therefore, a number of efficient approximation algorithms, called heuristics, has been developed in the last decades that can solve the traveling salesman problem

with acceptable accuracy in acceptable time, even with many nodes. However, a heuristic cannot guarantee that the optimal solution can actually be found.

Question 108

Level of difficulty: M

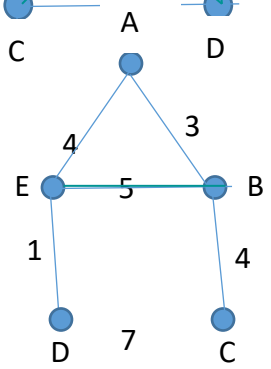
What is the solution to the traveling salesman problem, if the start is in node A?



Question 109

Level of difficulty: M

What is the solution to the traveling salesman problem, if the start is in node A?



Solution question 108

Solution hint: Hamiltonian cycle with minimum weighting

Short solution: e.g., ADBCA with total weight 11

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is ACBCA with total weight 11.

Solution question 109

Solution hint: Hamiltonian cycle with minimum weighting

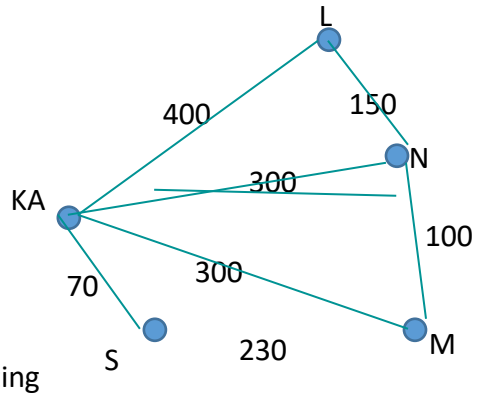
Short solution: AEDCBA with the total weight 19

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is AEDCBA with total weight 19.

Question 110

Level of difficulty: D

A traveling salesman must visit all cities.
Find the optimal circular route if it starts in KA.



Solution question 110

Solution hint: Hamiltonian cycle with minimum weighting

Short solution: KA S M N L KA with the total weight 950.

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is KA S M N L KA with total weight 950.

Lesson 8 - Trees

8.1 Properties of trees

Question 111

Level of difficulty: E

How many edges does a tree with n nodes have? Explain.

Question 112

Level of difficulty: E

A tree with n nodes has exactly n–1 edges. What can be derived from this?

Solution question 111

Solution hint: Successive picking of leaves

Short solution: n–1 edges, successive leaf picking technique.

Detailed solution: Each tree with n nodes has exactly n–1 edges. This can be traced using the technique of successive leaf picking. Let the number of nodes in the tree under consideration be n. If now the leaf and the corresponding edge are picked off, the result is a tree with one node and one edge less. However, the difference between the number of nodes and the number of edges in the old tree and the corresponding difference in the new tree remain the same. This process of picking off individual leaves is repeated until, after n–1 steps, a tree with one node has been created. This tree has no edge. This means that the difference between the number of nodes and the number of edges is equal to 1. Or, in other words, a tree with n nodes contains exactly n–1 edges.

Solution question 112

Solution hint: Familiarize yourself with the theory about trees.

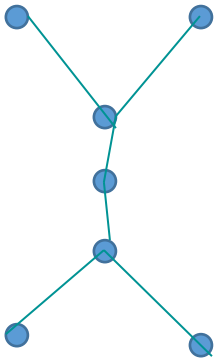
Solution: A graph is a tree if and only if it does not contain a closed walk, but adding exactly one new edge to the existing nodes creates a closed walk.

If a tree has a node of degree k, then it has at least k leaves. A connected graph is a tree exactly if removing any edge makes it no longer connected. A connected graph is a tree exactly if all its edges are bridges.

Question 113

Level of difficulty: M

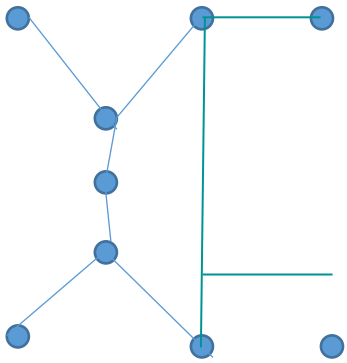
Decide if it is a tree and give reasons for your answer.



Question 114

Level of difficulty: M

Decide if it is a tree and give reasons for your answer.



Solution question 113

Solution hint: A graph with n nodes has n–1 edges.

Short solution: Yes

Detailed solution: The graph contains seven nodes and six edges, so it must be a tree.

Solution question 114

Solution hint: A graph with n nodes has n-1 edges.

Short solution: No

Detailed solution: The graph contains nine nodes and nine edges, so it cannot be a tree. Furthermore, it can be seen that it contains a closed walk.

Question 115

Level of difficulty: D

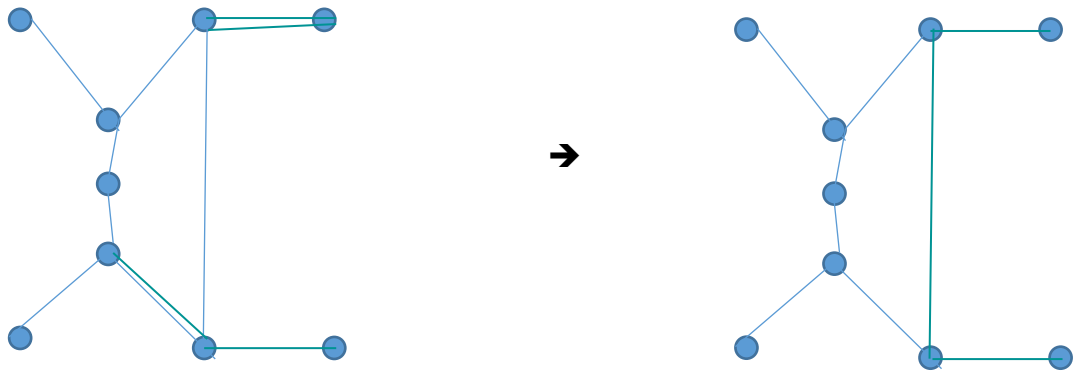
Is the following sentence true? If, from a connected graph with multiple edges, these multiple edges are removed, a tree is formed.

Solution question 115

Solution hint: Familiarize yourself with the requirements for a tree.

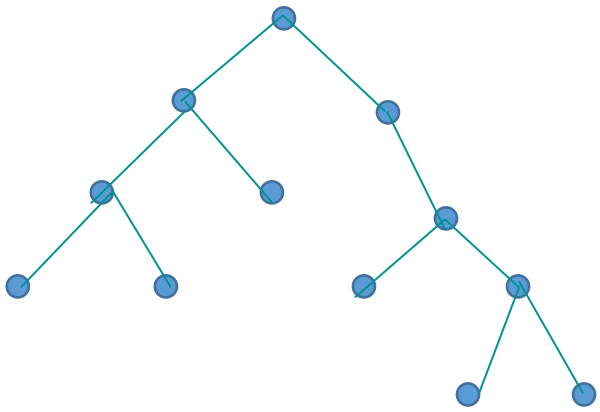
Short solution: Not true.

Detailed solution: If, for example, the multiple edges are removed from the graph below, the following graph is obtained:



The resulting graph is not a tree.

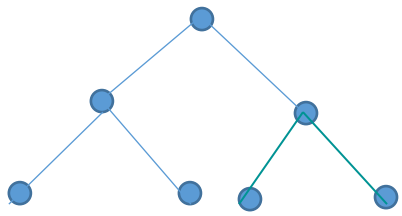
8.2 Rooted tree



Question 116

Level of difficulty: E

Given is the rooted tree of order 2. Decide whether it is complete and/or filled in. Justify.



Question 117

Level of difficulty: E

Given is the rooted tree of order 2. Decide whether it is complete and/or filled in. Justify.

Solution question 116

Solution hint: For a completed rooted tree, all inner nodes have the highest possible number of children, making the degree either equal to the order of the tree (inner node) or zero (leaf). A complete rooted tree has the highest possible number of nodes at each level if all inner nodes previously had the maximum number of children. Every complete rooted tree is also filled.

Short solution: Not filled in and therefore also not complete

Detailed solution: The given rooted tree is not completed because the node at depth 1 does not have enough children. Thus the rooted tree is also not complete.

Solution question 117

Solution hint: For a completed rooted tree, all inner nodes have the highest possible number of children, making the degree either equal to the order of the tree (inner node) or zero (leaf). A complete root tree has the highest possible number of nodes at each level if all inner nodes previously had the maximum number of children. Every complete rooted tree is also filled.

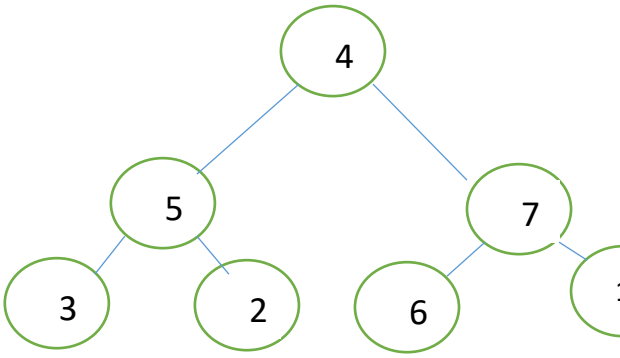
Short solution: Complete and filled in

Detailed solution: It is complete because there are two children at each level and so it is filled in.

Question 118

Level of difficulty: M

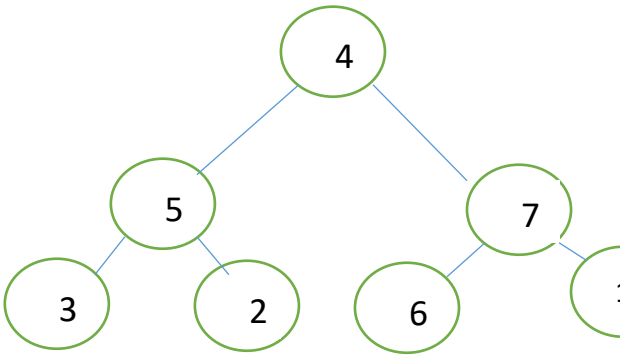
For the following binary search tree, enter the preorder sequence.



Question 119

Level of difficulty: M

For the following binary search tree, enter the inorder sequence.



Solution question 118

Solution hint: Familiarize yourself with the different traversing methods.

Short solution: 4 5 3 2 7 6 1

Detailed solution: First the root, then the left, then the right subtree: 4 5 3 2 7 6 1.

Solution question 119

Solution hint: Familiarize yourself with the different traversing methods.

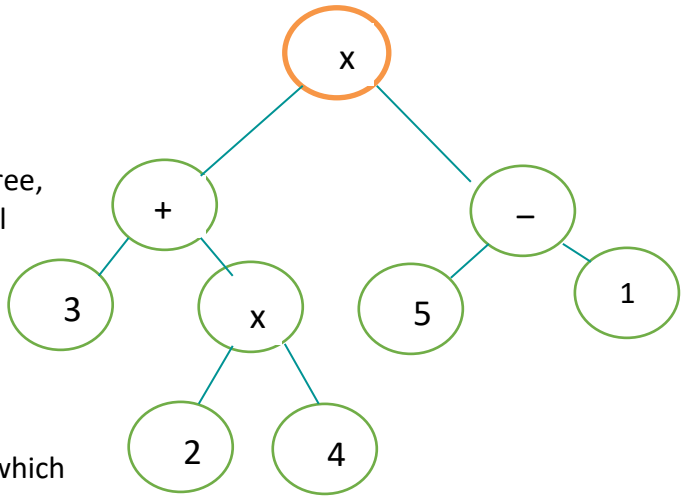
Short solution: 3 5 2 4 6 7 1

Detailed solution: First the left subtree, then the root, then the right subtree: 3 5 2 4 6 7 1.

Question 120

Level of difficulty: D

With a suitable traversing of the syntax tree, it is also possible to use the mathematical expression that is represented in the binary tree. For the given abstract syntax tree:



a) Which flow order is used?

b) What is the mathematical expression which is represented by the syntax tree?

Solution question 120

Solution hint: Familiarize yourself with the traversal.

Short solution: a) Inorder sequence b) 44

Detailed solution: a) The pass order by inorder sequence leads to the mathematical expression, b) $(3 + (2 \cdot 4)) \cdot (5–1) = 44$.

8.3 Spanning tree

Question 121

Level of difficulty: E

What is meant by a tree? Explain how to obtain a tree from any graph that is not one.

Question 122

Level of difficulty: E

How many spanning trees does the complete graph K_5 have?

Solution question 121

Solution hint: Familiarize yourself with the theory of the spanning tree. A tree must not have closed walks.

Short solution: A tree containing all the nodes of a graph; find a closed walk from a graph that is not a tree and eliminate it by removing an edge, and so on.

Detailed solution: A tree containing all nodes of a graph is a spanning tree of the graph. This is created when a closed edge move is picked from any graph that is not a tree and an edge is removed from it, eliminating the closed walk. If there are other closed walks in the graph, another edge is removed from them, further reducing the number of closed walks. In this way, edges are removed until there are no more closed walks and the graph has become a "thinned" tree.

Solution question 122

Solution hint: Cayley's formula

Short solution: 125

Detailed solution: The formula of Cayley (1889) gives an upper limit. It says: A complete graph K_n has n^{n-2} spanning trees. Thus K_5 has 125 spanning trees.

Question 123

Level of difficulty: M

Explain the breadth-first search algorithm for finding spanning trees.

Question 124

Level of difficulty: M

Explain the depth-first search algorithm for finding spanning trees.

Solution question 123

Solution hint: The algorithm traverses all possible walks of length 1 from a node.

Short solution: The algorithm traverses all possible edge paths with length 1 from a node. From the newly reached nodes, it again follows walks of length 1 to unreached nodes. This is continued until there are no more unreached nodes. Since the graph is connected, the algorithm finds a spanning tree after finitely many steps.

Detailed solution:

- Step 1: Set E_0 as the empty set and define two lists L_1 and L_2 also as empty lists. An arbitrary starting node v_1 is chosen and written to lists L_1 and L_2.
- Step 2: If L_1 is empty, then stop and $T = (V, E_0)$ is a spanning tree. Otherwise, take the first node v from L_1.
- Step 3: If all adjacent nodes of v are in L_2, then v is deleted from L_1. Otherwise, choose the "smallest" adjacent node w that is not in L_2 and write it to the end of the lists L_1 and L_2. The edge $\{v, w\}$ is added to E_0.
- Step 4: Go to step 2.

Solution question 124

Solution hint: From the start node, find a walk that is as long as possible.

Short solution: An attempt is made to find the longest possible walk from the start node that leads as far as possible from the start node, i.e., "deep" into the tree. If the walk can no longer be extended, since it would otherwise be closed, the algorithm returns along the walk until it reaches a node from which another unvisited node can be reached. With this node it starts again into the "depth" of the graph. When no more unvisited nodes are left, the algorithm stops and a spanning tree is the result.

Detailed solution:

- Step 1: Set E_0 as the empty set and define two lists L_1 and L_2 as empty lists. An arbitrary starting node v_1 should be chosen and written to the lists L_1 and L_2.
- Step 2: If L_1 is empty, then stop; $T = (V, E_0)$ is a spanning tree. Otherwise, take the first node v from L_1.
- Step 3: If all adjacent nodes of v are in L_2, then v is deleted from L_1. Otherwise, the "smallest" adjacent node w that is not in L_2 is chosen and written to the beginning of lists L_1 and L_2. The edge $\{v, w\}$ is added to E_0.
- Step 4: Go to step 2.

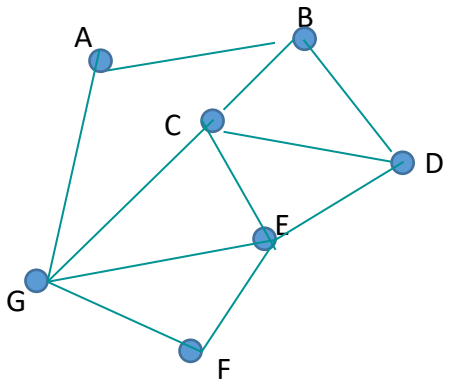
Question 125

Level of difficulty: D

Give a spanning tree for the graph in each case.

Start with node A.

a) By using the breadth-first search
b) By using the depth-first search



Solution question 125

Solution hint: a) Starting from a node walk all possible walks of length 1 b) start from the start node and find a walk that is as long as possible

Solution: a) {{A, B}, {A, G}, {B, C}, {B, D}, {G, E}, {G, F}, b) {A, B}, {B, D}, {D, C}, {C, E}, {E, F}, {F, G}}; b) several solutions are possible

8.4 Minimum spanning tree

Question 126

Level of difficulty: E

Explain Kruskal's algorithm.

Question 127

Level of difficulty: E

Explain the problem of the most convenient connection. What else is this problem called? What is meant by a minimum spanning tree? Give an example.

Solution question 126

Solution hint: "Greedily" reach for one of the cheapest edges

Short solution: Kruskal's algorithm (1956) belongs to the greedy algorithms. Here, one always greedily grabs one of the cheapest edges.

Detailed solution: For a connected weighted graph with n nodes, Kruskal's algorithm determines a minimum spanning tree T:

- Step 1: Set T: = Ø, i: = 0.
- Step 2: Among the edges not yet visited, visit the one with the smallest weight. If it closes an edge in T, discard it; otherwise, add it to T and set i: = i + 1.
- Step 3: If i = n–1, then stop, otherwise continue with step 2.

Solution question 127

Solution hint: Minimum spanning tree

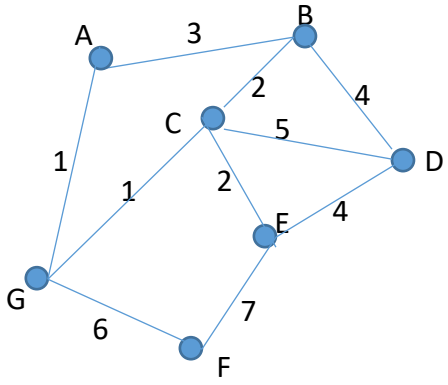
Short solution: Minimum spanning tree, connector problem, railway network with minimum total weight

Detailed solution: The problem of the most favorable connection (connector problem) is represented graph theoretically by a minimum spanning tree. A spanning tree for a weighted graph whose total weight is minimum is called minimum spanning tree T (also called minimum spanning tree or MST). Example: Railway network; all cities are to be reached with minimum total weight.

Question 128

Level of difficulty: M

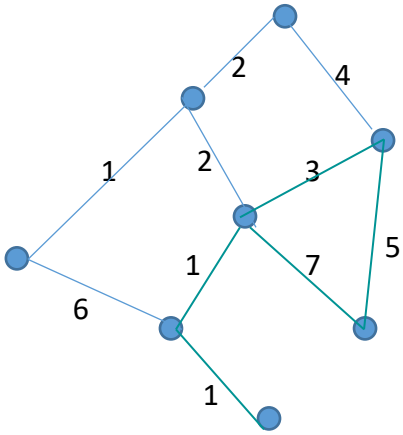
New roads are to be built between cities. The construction costs are given in millions of euros. As this sum is not available at that present moment, it is planned that each city can be reached by every other city on one of the new roads, but by only one road at a time. The construction costs should also be kept as low as possible. Therefore, a minimum spanning tree of the graph is searched.



Question 129

Level of difficulty: M

In this graph, a minimum spanning tree should be found. Specify the total weight.

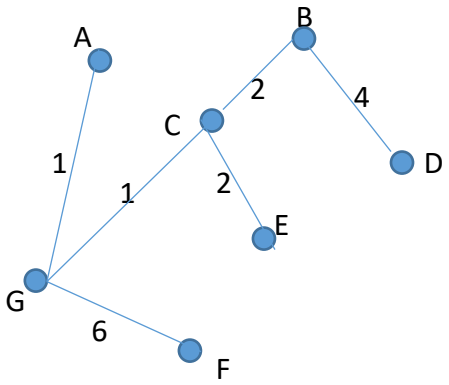


Solution question 128

Solution hint: Kruskal's algorithm

Short solution: The total weight is 16.

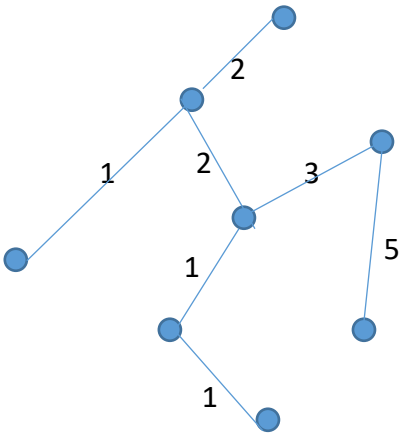
Detailed solution: To find this, Kruskal's algorithm is used and for this all weights are noted in ascending order as a list: 1 1 2 2 3 4 4 5 6 7. Then, the most favorable road is built first, followed by the next most favorable and so on. With equal weights it does not matter which one is chosen. Only care must be taken that no closed walks will be created. If all cities have a road connection, the task is solved. Total weight: 16.



Solution question 129

Solution hint: A tree containing all nodes and where the total weight is minimal.

Short solution: The total weight is 15.



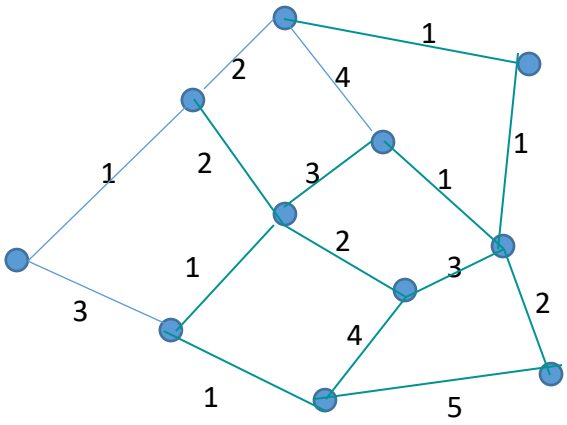
Detailed solution: 1 1 1 2 2 3 4 5 6 7

Total weight: 15.

Task 130

Level of difficulty: D

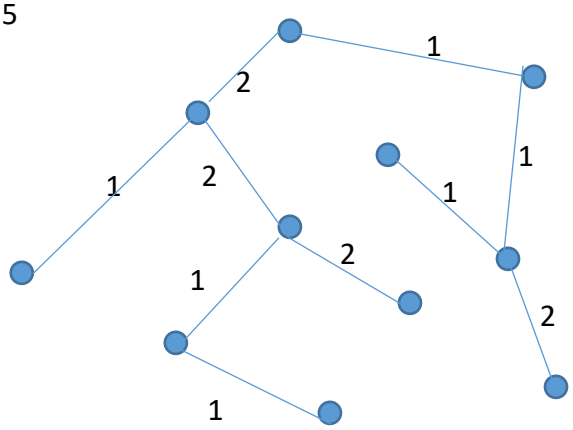
In this graph, a minimum spanning tree can be found. Enter the total weight.



Solution Task 130

Solution hint: A tree containing all nodes and where the total weight is minimal

Short solution: The total weight is 14.

Detailed solution: 1 1 1 1 1 1 2 2 2 2 3 3 3 4 4 5

Total weight: 14.